

SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

|                  |                                                           |
|------------------|-----------------------------------------------------------|
| Sector           | Residential                                               |
| Building type    | Multi-Unit                                                |
| Building subtype | Duplex - Detached, conditioned basement (with main floor) |
| Vintage          | pre-1945                                                  |

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation and performance analysis.

BUILDING CHARACTERISTICS

|                     |                                   |
|---------------------|-----------------------------------|
| Floor Area [m²]     | 318                               |
| Climate zone        | 6A                                |
| HVAC SYSTEM         |                                   |
| Cooling system      | Air-Source Heat Pump (Mini-Split) |
| Heating system      | Electric Baseboards               |
| Ventilation system  | /                                 |
| Typical COP cooling | 2.50                              |

INSULATION & AIR TIGHTNESS

|                               |        |
|-------------------------------|--------|
| Exterior walls [W/m²/K]       | 0.6    |
| Interior walls [W/m²/K]       | 0      |
| Roof [W/m²/K]                 | 0.45   |
| Slabs [W/m²/K]                | 4.16   |
| Infiltration (ELA @4Pa) [cm²] | 429.21 |

GLAZING

|                  |      |
|------------------|------|
| Glazing [W/m²/K] | 2.94 |
| SHGC             | 0.6  |

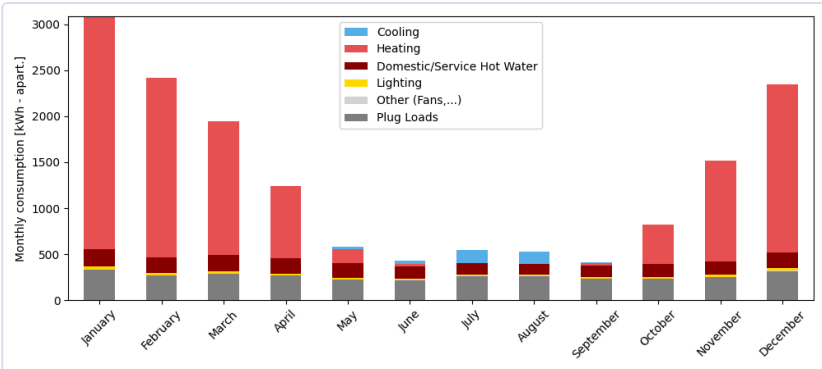
WINDOW-TO-WALL RATIO

|             |     |
|-------------|-----|
| Average [%] | 4.3 |
|-------------|-----|

ANNUAL END USES — ELECTRICITY [kWh]

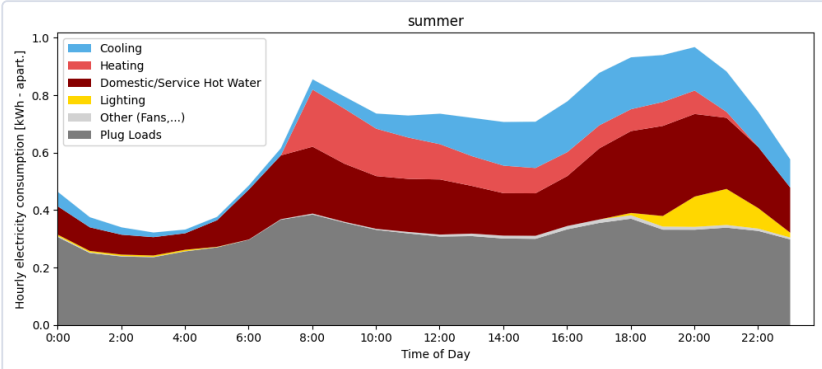
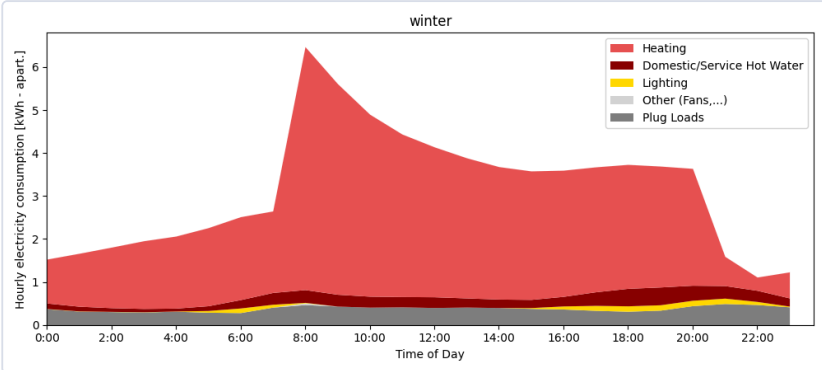
| END USE                    | [kWh/year per apart.] |
|----------------------------|-----------------------|
| Heating                    | 10285                 |
| Cooling                    | 354                   |
| Domestic/Service Hot Water | 1808                  |
| Lighting                   | 258                   |
| Plug Loads                 | 3138                  |
| Other (Fans,...)           | 31                    |
| Total                      | 15874                 |

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption breakdown by end use, showing the contribution of heating, cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout the year.

AVERAGE DAILY PROFILE BY USAGE



Average daily consumption profile for winter and summer, showing the contribution of cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout an average winter and summer day.

ENERGY USE INTENSITY

50

kWh/m²/yr

TOTAL CONSUMPTION

15875

kWh/yr

ELECTRICITY

15874

kWh/yr

NATURAL GAS

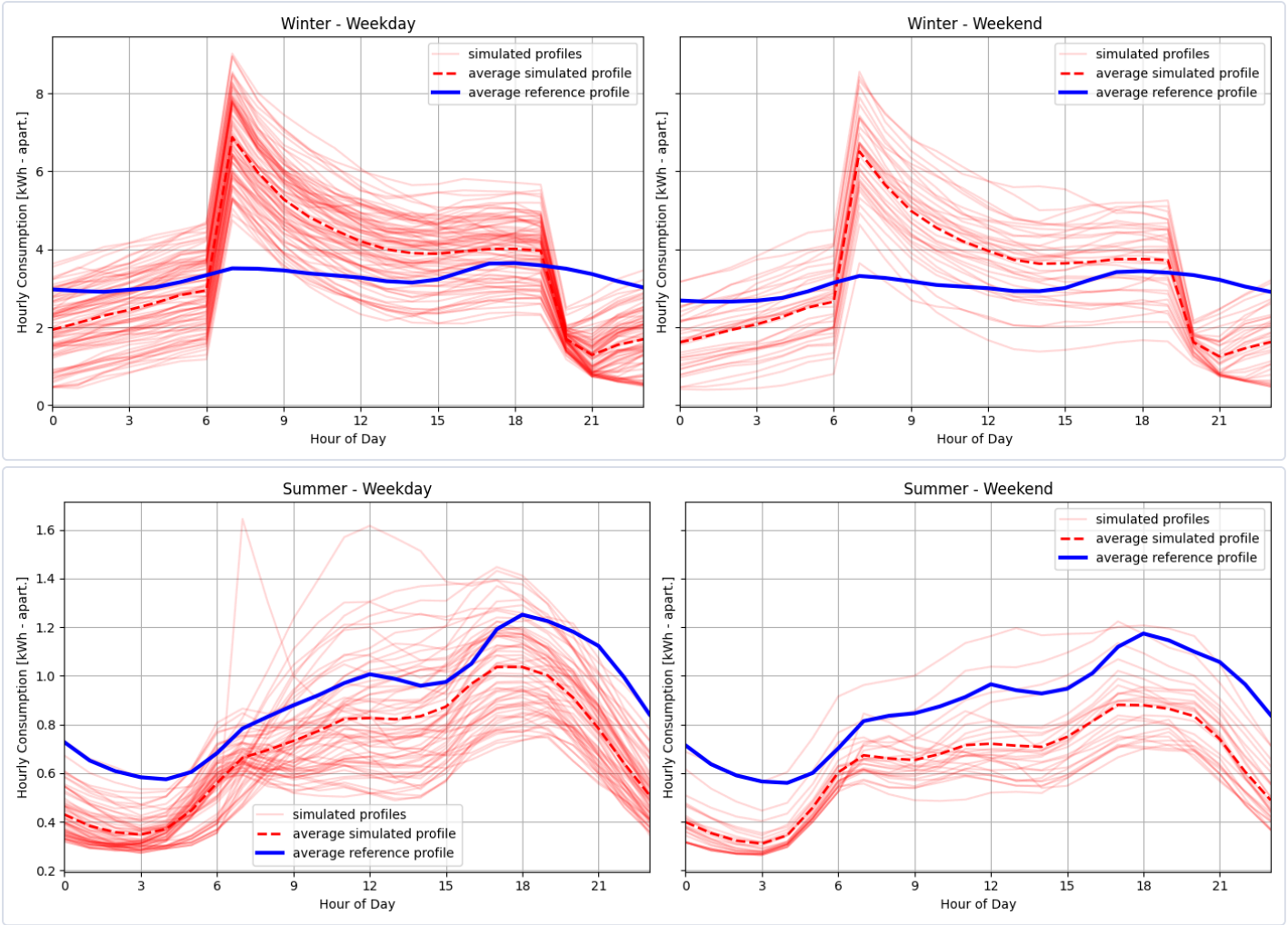
0

kWh/yr

DONNÉES DE RÉFÉRENCE

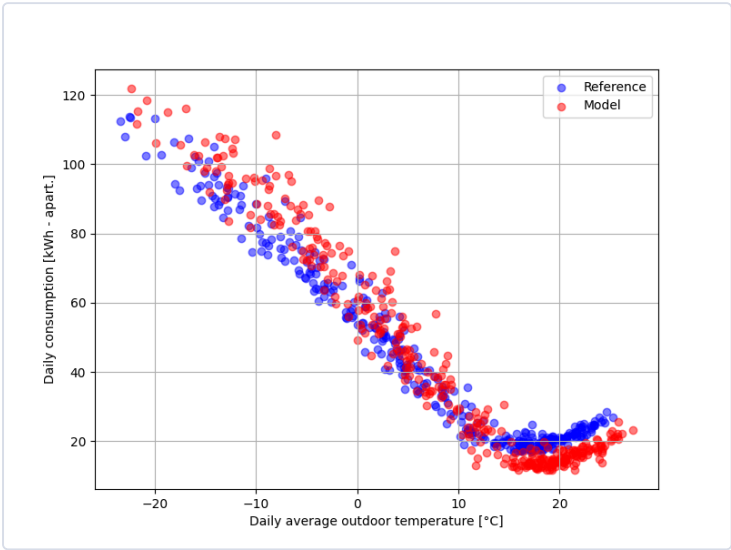
The reference data used for the comparison come from a dataset of more than 60,000 residential electricity meters, combined with metadata. Filters were applied to ensure consistency with the building type and subtype of this card, as well as to exclude end uses such as pool, spa, and electric vehicle charging.

DAILY PROFILE



Daily consumption profiles for summer and winter comparing simulated hourly energy use with the reference data. Red lines represent individual days from the model for the given period, the red dashed line indicates the average simulated profile, and the blue line shows the average profile of the reference data. The figures on the left represent weekdays, while the figures on the right represents weekends.

DAILY CONSUMPTION VS. DAILY AVERAGE OUTDOOR TEMPERATURE



Comparison between modeled daily consumption and reference consumption as a function of outdoor temperature, for all days of the reference year.

COMPARISON TABLE — KEY INDICATORS

| INDICATOR                            | MODEL  | REFERENCE | DELTA |
|--------------------------------------|--------|-----------|-------|
| Yearly electricity consumption [kWh] | 15875  | 16138     | -264  |
| Baseload consumption [kWh/day]       | 14.0   | 19.4      | -5.4  |
| Electric heating slope [W/K]         | -123.2 | -111.0    | -12.2 |
| Cooling slope [W/K]                  | 48.2   | 55.0      | -6.8  |
| Winter peak demand - AM [kW]         | 9.0    | 5.2       | 3.8   |
| Winter time of peak - AM             | 7.0    | 9.0       | -2.0  |
| Winter peak demand - PM [kW]         | 6.1    | 5.3       | 0.8   |
| Winter time of peak - PM             | 12.0   | 18.0      | -6.0  |